

## **Response Letter with Annotations**

Bail, C. A., Argyle, L. P., Brown, T. W., Bumpus, J. P., Chen, H., Hunzaker, M. B. F., Lee, J., Mann, M., Merhout, F., & Volfovsky, A. (2018). Exposure to opposing views on social media can increase political polarization. *Proceedings of the National Academy of Sciences*, 115(37), 9216–9221. <https://doi.org/10.1073/pnas.1804840115>

### **Annotations**

#### **1. Purposes**

##### **a. Motivation:**

- The motivation to produce this study was due to growing concerns about social media echo chambers that may be driving growth in partisan divides in the United States.

##### **b. Specific Rationale (several bullet points)**

- Political polarization in the United States has increasingly become more divisive, especially on controversial issues such as inequality, gun control, and immigration.
- These partisan divides are often accredited to and blamed on “echo chambers” and selective exposure of political information on social media.
  1. This is especially concerning when you consider how the majority of Americans go on social media site at least once a day and use social media as their primary news medium
- The attribution of social network formations effect on political attitudes and polarization has been challenging to establish given how strenuous it is to prove whether social networks shape political attitudes, or if the inverse is true.

##### **c. Research Questions and/or Hypotheses**

- Identify IV, DV, mediator, moderator
  1. IV: Exposure to opposing-party Twitter content
  2. DV:
    - a. Political polarization/attitude
      - i. Level of attachment to political party and
      - ii. Level of interest in current events
  3. Moderators:
    - a. Party identification (Democrat vs. Republican)
    - b. Exposure intensity/ Treatment compliance
  4. Mediator: N/A
  5. Control:
    - a. Demographic variables
      - i. Age, income, education, gender, race, and geographic region
    - b. Respondents’ pretreatment liberal/conservative scale score
      - i. Done to mitigate the influence of period effects
    - c. Network-based ideological scoring - Twitter

- i. Percentage of people who the participant follows on Twitter that share party identification pre-treatment
  - d. Network-based ideological scoring - Offline
    - i. Percentage of people in the participants offline network who share their party identification
  - e. Numeric measure participant following
    - i. Models continuously measure of the number of people the participant follows to account for the likelihood of bot message exposure
- Identify specific relationships among variables
  1. Among the variables, exposure to opposing political views within Twitter content was predicted to decrease political polarization.
  2. However, exposure to opposing political views was also predicted to increase backfiring effects which increase political polarization.
  3. Finally, backfiring effects among the exposure of opposing political views was predicted to happen more for those with conservative political views.
- Hypotheses- preregistered
  1. **H1:** Disrupting selective exposure to partisan information *will decrease* political polarization because of intergroup contact effects
  2. **H2:** Exposure to those with opposing political views *will create backfire effects* that exacerbate political polarization
  3. **H3:** Backfire effects will be *more likely* to occur among conservatives than liberals.

## 2. Method

- a. Research Design
  - The study used a large-scale field experiment with U.S. Twitter users, combining “ostensibly unrelated” online surveys, randomized assignment to Twitter-bot exposure, and behavioral compliance checks to measure ideological change.
- b. Sample (representative sample?)
  - Pretreatment (baseline) survey:
    1. Total N = 1,652 (901 Democrats, 751 Republicans)
  - Analyzed experimental sample (those with pre and post data):
    1. Total N = 1,239 (697 Democrats, 542 Republicans)
  - Survey Firm: YouGov
    1. Used U.S. census sampling frames to have representativeness in participants
- c. Measures
  - Primary outcome: political polarization/ attitude
    1. A 10-item policy attitude scale: respondents rated agreement with statements on social issues on a 7-point scale ranging from “strongly disagree” to “strongly agree.”; items were combined into a liberal-conservative scale
      - a. Example statements on social issue:

- i. "Stricter environmental laws and regulations cost too many jobs and hurt the economy."
    - ii. "Immigrants today strengthen our country because of their hard work and talents."
  - 2. Measured pretreatment and one month later to capture change
- Other measures
  - 1. Social and conventional media use
  - 2. Demographics
  - 3. Twitter behavior (who they follow) via reported Twitter IDs
    - a. Digital trace data
  - 4. Treatment compliance via weekly quizzes on bot tweet content and identifying a daily animal image
- d. Analysis
  - Multivariate regressions

### 3. Results and Conclusions

#### a. Answers to RQs and Hypotheses

- **H1: Not Supported**
  - 1. Disrupting selective exposure to partisan information *increased* political polarization
- **H2: Partial Support**
  - 1. Exposure to those with opposing political views *did create backfire effects* that exacerbated political polarization, but this mainly occurred with Republican participants
  - 2. Democrat participants had *slightly more* liberal attitudes after exposure to opposing political views, but this effect wasn't statistically significant
- **H3: Supported**
  - 1. Backfire effects did *occur more* among conservatives than liberals.

#### b. Include Key Figures/Tables

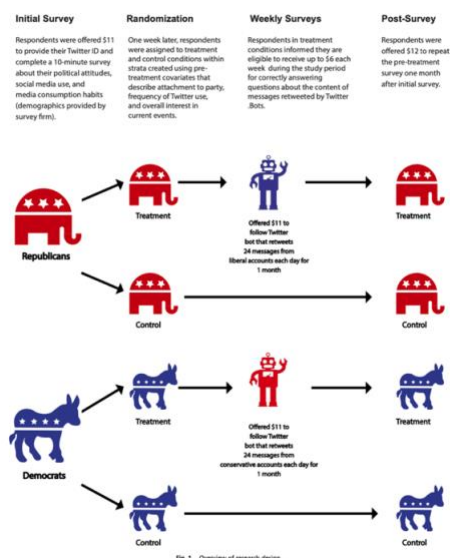


Fig. 1. Overview of research design.

Figure 1

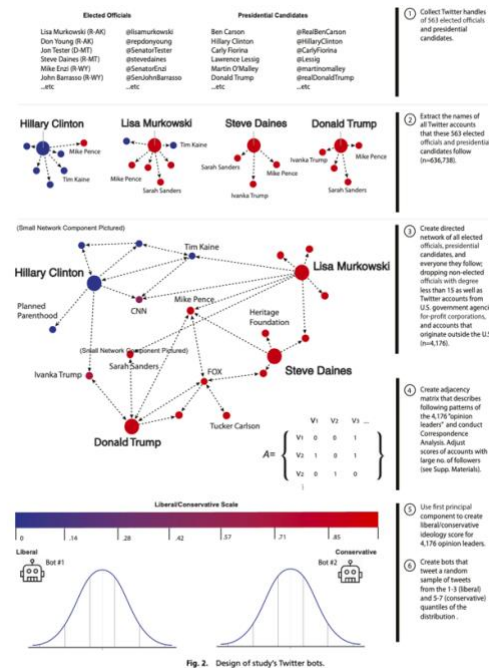


Figure 2

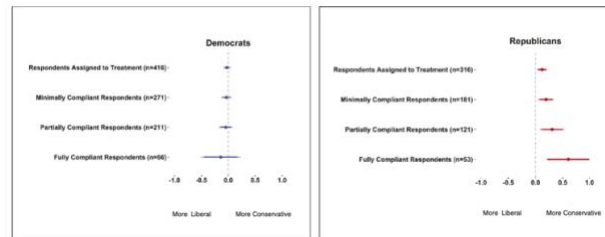


Fig. 3. Effect of following Twitter bots that retweet messages by elected officials, organizations, and opinion leaders with opposing political ideologies for 1 mo, on a seven-point liberal/conservative scale where larger values indicate more conservative opinions about social policy issues, for experiments with Democrats ( $n = 697$ ) and Republicans ( $n = 542$ ). Models predict posttreatment liberal/conservative scale score and control for pretreatment score on this scale as well as 12 other covariates described in *SI Appendix*. Circles describe unstandardized point estimates, and bars describe 90% and 95% confidence intervals. "Respondents Assigned to Treatment" describes the ITT effect for Democrats (ITT =  $-0.02$ ,  $t = -0.76$ ,  $p = 0.45$ ,  $n = 416$ ) and Republicans (ITT =  $0.12$ ,  $t = 2.68$ ,  $p = 0.008$ ,  $n = 316$ ). "Minimally-Compliant Respondents" describes the CACE for respondents who followed one of the study's bots for Democrats (CACE =  $-0.04$ ,  $t = -0.75$ ,  $p = 0.45$ ,  $n$  of compliant respondents = 271) and Republicans (CACE =  $0.19$ ,  $t = 2.73$ ,  $p < 0.007$ ,  $n$  of compliant respondents = 181). "Partially-Compliant Respondents" describes the CACE for respondents who correctly answered at least one question, but not all questions, about the content of a bot's tweets during weekly surveys throughout the study period for Democrats (CACE =  $-0.05$ ,  $t = -0.75$ ,  $p = 0.45$ ,  $n$  of compliant respondents = 211) and Republicans (CACE =  $0.31$ ,  $t = 2.73$ ,  $p < 0.007$ ,  $n$  of compliant respondents = 121). "Fully-Compliant Respondents" describes the CACE for respondents who answered all questions about the content of the bot's tweets correctly for Democrats (CACE =  $-0.14$ ,  $t = -0.75$ ,  $p = 0.46$ ,  $n$  of compliant respondents = 66) and Republicans (CACE =  $0.60$ ,  $t = 2.53$ ,  $p < 0.01$ ,  $n$  of compliant respondents = 53). Although treated Democrats exhibited slightly more liberal attitudes posttreatment that increase in size with level of compliance, none of these effects were statistically significant. In contrast, treated Republicans exhibited substantially more conservative views posttreatment that increase in size with level of compliance, and these effects are highly significant.

Figure 3

## c. Authors' Conclusions

- This study found that exposure to opposing political views does not have a depolarizing effect on participants. Instead, it caused backfiring effects among Republican participants which resulted in substantially more conservative attitudes in posttreatment. Among Democrat participants, there was only slight shifts towards liberal leaning attitudes which were ultimately insignificant on a statistical level.

## 4. Discussion

### a. Implications

- This study is methodologically innovative as it allowed researchers to collect and connect social media and politics with high granularity. This plus developing techniques for measuring exposure to content, mitigating causal

interference, and verifying survey responses with behavioural data is an advancement in field of computational social science.

- This study have implications for those interested in reducing political polarization. Findings suggest that efforts to introduce politically polarized audiences to opposing political views on social media is not only an ineffective strategy, but a counterproductive one that can cause further polarization especially among Republicans.

**b. Limitations**

- These findings are not generalizable to the entire United States (U.S.) populus because the majority of U.S. citizens don't use Twitter
- These findings are not generalizable to other countries or social media platforms, and it is important to understand if these findings occur in other social network, cultural, and political contexts
- This study excluded independents and those who infrequently use Twitter from their analysis, this indicates that these individuals could have had a differing reaction to exposure of opposing political views.
- It is unknown what specific mechanism of messaging within the randomly generated Tweets caused backfiring effects.
  1. The intervention in this study only exposed participants to high-profile elite accounts with opposing political ideologies. While context was randomly generated with the bot these elites are predominately more polarized than the average citizen. Backfiring effects seen in this study could have occurred due to antielite bias instead of the political messaging on its own.

**c. Future Directions**

- Future studies are needed to determine whether these findings can be replicate in different global populations or within varied social media platform settings.
- Future studies are needed to identify the precise causal pathways that create backfiring effects.

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Response Letter (~600 words)

- **Your assessment of the study**

Overall, this study was a fascinating and substantive read. The first thing that struck me was the development and advancement of the methods and how incredibly innovative they were. To utilize an “ostensibly unrelated” survey design to ensure that participants were unaware of the politicalization focus of the study was well strategized; but to further combine this survey design with an online field experiment was creative. Due to this research design, results from this study have great balance between internal and external validity. This design not only improves upon the limitations within field experiment methods (i.e. nonreactivity, construct validity, internal validity and control over treatment exposure) but it also improves the limitations within survey design (i.e. self-report bias and external validity and realism). Finally, and as the authors point out, this methodological approach is a major progression in computational social science research.

Beyond the methodological strong points, the narrative and figures presented throughout the study were clear and concise. From the beginning of the paper, the importance of this study and why it needs to be done in this current moment is explicit: based on rising concerns among Americans that social media contributes to political echo-chambers and amplifies polarization, it is important to discern if injecting opposing views into said echo chambers will increase or decrease individual polarization. This “so what?” hooks readers into the study and keeps them grounded throughout the article. From there the reasoning and explanation for hypotheses, methods, and findings flows together and it always goes back to answering this “so what?”.

Furthermore, the figures throughout this article were helpful in the understanding of this piece. The figures created were well-designed and fully demonstrated the research design, design of Twitter bots, and results of the study through text and graphic visuals. Especially for those new to this use of computational methods within social science, these figures actively support comprehension.

Finally, one of my biggest takeaways from this study was how fascinating and practically applicable the findings were. Especially in our current political climate, it is incredibly valuable to know that inserting opposing views to create a more balanced social media stream can actually do more harm than good. This furthers knowledge surrounding political polarization and demonstrates that this “simple fix” is not a way to mitigate it. These findings necessitate building knowledge not only around depolarization, but understanding the mechanisms, messages, and additional factors that create backfiring effects.

- **Weaknesses**

Even though this study had several strong suits, there are weaknesses that need to be addressed. Besides the various limitations listed within this article, one major issue I noticed from the jump was how the competing hypotheses are presented. While competing hypotheses are an acceptable practice within research, it is important that they are clearly communicated as competing hypotheses and that the theoretical rational for having competing hypotheses is communicated. Otherwise, you run the risk of catching readers off guard and taking them out of the narrative of your study by questioning what is to be expected. However, this study did pre-register their hypotheses and provided ample amounts of research and theory to justify each hypothesis. If these hypotheses were presented as

competing based on various theoretical alternatives, then it would be clearly justified and cease potential questioning.

Another major weakness stems off the limitation in identifying what mechanism caused backfiring effects among Republicans. Within this study the only aspect of the retweets that is explained is that political elites initially wrote them. Unfortunately, we don't know what the contents of these tweets actually contained. Even in the SI Appendix, which is mentioned several times throughout the study, the only acknowledgement of what tweets contained was the pictures of nature used in the first few days of the study to mask participants to the purpose of the study. Explanation here of what tweets often contained would've supported the clarity of this study and provide potential avenues for secondary analysis. Several aspects of this study, like measures, control variables, and analysis, had to be explained in further depth in the SI Appendix because there was likely a limited word count for this journal. Even with these additional weaknesses, this study is exceedingly strong in its methods and findings.

- **Other notes**

An additional point that I've been thinking about is how these findings can build off of mass communication theories like hostile media effects. Within hostile media effects research, it is shown that being forced to read neutral newspapers results in higher perceptions of bias between groups, whereas selective exposure to neutral newspaper articles did not result differ statistically in bias perceptions (Mende, 2008). Within the context of this study, I wonder if hostile media effects apply or if another mass communication theory can provide further explanation to the results.

Mende, A. M. (2008). Testing the hostile media effect under selective exposure. In International Communication Association Conference, Montreal, Canada.